

Lab 1: Music Genre Classification

Name

Germán Puerto Rodríguez

Link

<https://colab.research.google.com/drive/14oUX0FyKxS5RaD7NFq5E2GYdNDMrqwmE?usp=sharing>

Results

Comparative Analysis

The three tasks demonstrate a clear trade-off between **Feature Source Complexity**, **Sample Size**, and the resulting **Accuracy/Robustness** achieved by the models.

Metric	Task 1: Task 2: Task 3:
	Essentia (Full Audio) Features_30_Segs Features_3_Secs
Sample Size (N)	≈1,000 ≈1,000 ≈10,000
Original Features (D)	243 (High-dim) 58 (Mid-dim) 58 (Mid-dim)
Peak Accuracy (Best Model)	0.8894 (SVC RBF) 0.8200 (ETC) 0.9209 (KNN)
Best Robustness (Min OF Gap)	7.99% (SVC Lin) 6.40% (Log Reg) 1.33% (Log Reg)
Primary Challenge	Overfitting (High Variance) Overfitting (High Variance) Accuracy Maximization
Winning Strategy	RFECV → PCA Data Cleaning + Feature Selection Large Sample Size

1. The Dominance of Sample Size (Task 3 vs. Tasks 1 & 2)

The shift from $\approx 1,000$ samples (Tasks 1 & 2) to $\approx 10,000$ samples (Task 3) is the most significant result.

- **Accuracy Ceiling Raised:** The peak accuracy jumped from the high 80s (0.8894 in Task 1) to over 92% in Task 3.
- **Overfitting Solved:** Task 3 effectively eliminated the overfitting problem. The **minimum Overfitting Gap** plummeted from the best of 7.99% (Task 1) to an astonishing 1.33% (Task 3). The average model variance in Task 3 became acceptable without aggressive feature tuning.
- **Model Shift:** In Tasks 1 & 2, complex, regularized models like **SVC** were necessary. In Task 3, the simple, non-parametric **K-Nearest Neighbors (KNN)** became the champion, benefiting directly from the abundance of high-quality data points.

2. Feature Source and Dimensionality (Task 1 vs. Task 2)

Comparing the two $N \approx 1,000$ tasks isolates the effect of the **feature source**.

- **Essentia (Task 1):** The initial 243 features led to higher overall accuracy (**0.8894**) but required much more complex preprocessing (**RFECV $\rightarrow$ PCA**) to manage the extreme dimensionality and control overfitting. The higher feature count provided more raw predictive power, assuming it was successfully curated.
- **30-Second Descriptors (Task 2):** The 58 features were less predictive (Peak Acc. **0.8200**) but were more easily stabilized. The best strategy focused heavily on **data cleaning** and simple feature filtering to bring the high initial variance under control.

Conclusion on Low N: high-dimensional feature set (like Essentia) can achieve higher accuracy, but **only** with advanced dimensionality reduction techniques.

3. Final Strategic Conclusions

Strategy	Recommended Task	Reason
Best Accuracy	Task 3 (3-Second Clips)	92.09% acc. Large size is the greatest asset. KNN model with all 58 features.
Most Robust/Stable	Task 3 (3-Second Clips)	Lowest variance (1.33% OF Gap). Logistic Regression model for extreme stability.
Best Low-Data Accuracy	Task 1 (Essentia)	88.94% accuracy. Combining IQR Capping and RFECV → PCA .

The analysis conclusively proves that **increasing the number of short audio samples (Task 3)** is the most effective approach for high-performance and robust music genre classification.

Results Summary

Essentia Dataset

The exhaustive analysis identified that data cleaning and dimensionality reduction were critical to achieving reliable performance on the N=993 Essentia dataset.

1. Optimal Strategy and Performance

Metric	Feature Set	Model	Result
Peak Accuracy	RFECV → PCA (47 Components)	SVC (RBF Kernel)	0.8894
Best Robustness	RFECV → PCA (47 Components)	SVC (Linear)	0.8593 Acc. (with 7.99% OF Gap)

2. Key Findings

- **Data Cleaning is Essential:** Applying **IQR Outlier Capping** provided an immediate ≈2.5% accuracy improvement over the raw baseline.
- **Overfitting Management:** All feature reduction techniques (RFECV, PCA, CFS) successfully mitigated the extreme baseline overfitting (was ≈20%), bringing the most stable models into the desirable 7%–10% variance range.
- **Algorithm Dominance:** The **SVC** family consistently outperformed all other models, including complex ensembles like Random Forest and Extra Trees.

3. Conclusion

The recommended strategy for deployment uses the **RFECV → PCA (47 components)** feature set. While the SVC (RBF Kernel) yields the highest raw accuracy (88.94%), the **SVC (Linear)** model is the most reliable choice, offering a strong 85.93% accuracy with the best stability (7.99% gap).

Features 30 Seconds Dataset

1. Optimal Strategy and Performance

Metric	Feature Set	Model	Result
Peak Accuracy	RFECV (48 Features)	Extra Trees Classifier (ETC)	0.8200
Best Robustness	RFECV → RF Top 40	Logistic Regression	0.7200 Acc. (with 6.40% OF Gap)

2. Key Insights

- **Preprocessing is Essential:** Initial **IQR Outlier Capping** provided the most critical stability gain, drastically reducing the overfitting gap for linear models (e.g., Log Reg gap reduced by 8.44%).
- **Accuracy vs. Stability:** The **ETC** achieved the accuracy ceiling, but the **SVC (Linear)** and **Logistic Regression** models, paired with reduced feature sets, proved far more robust, achieving the only sub-10% overfitting gaps.
- **Feature Strategy:** The **RFECV**-selected feature set (48 features) provided the highest quality features for non-linear ensemble models.

3. Conclusion

For a balanced production deployment where stability is prioritized over peak raw score, the **SVC (Linear) model** trained on the **PCA (90% / 22 components)** feature set is recommended, offering reliable 73.50% accuracy with a low 9.61% variance.

Features 3 Seconds Dataset

For task 3, the most complex models (Random Forest, SVM) were not included in the evaluation because their complexity makes the entire grid search or cross validation pipeline for a dataset 10 times larger take a long time to process, and we are looking for models that have a good relationship between accuracy and overfitting but also training time.

Furthermore, due to the features of the problem, the excessive nonlinear complexity of models like these is not necessary.

1. Optimal Strategy and Performance

Metric	Feature Set	Model	Result
Peak Accuracy	Full 58 Features (Exp 1)	K-Nearest Neighbors (KNN)	0.9209
Best Robustness	PCA 30 Components (Exp 3)	Logistic Regression	1.33% OF Gap (0.6682 Acc.)

2. Key Insights

- **Sample Size Solves Overfitting:** The massive increase in training data (N=9990) fundamentally resolved the high variance problem prevalent in the smaller dataset. The **Logistic Regression** model's gap dropped from 20.40% to 2.50%.
- **KNN Dominance:** KNN emerged as the superior non-linear classifier, achieving a near-perfect 92.09% accuracy, validating the quality of the 3-second audio features.
- **Feature Retention:** PCA was detrimental to accuracy, dropping the peak score by 6%. The models perform best when retaining the **full 58 features**.
- **Capping Redundancy:** IQR Outlier Capping offered only a marginal 0.1% gain, confirming its low importance with large datasets.

3. Recommendation

The definitive recommendation is to use the **K-Nearest Neighbors (KNN) model** with the **full 58 features**. This strategy delivers state-of-the-art accuracy (92.09%) while maintaining a highly acceptable stability level (OF Gap $\approx$ 9.5%).

Results Analysis

Essentia Dataset

The machine learning experiments conducted on the Essentia dataset (993 rows, 243 features) demonstrate a clear relationship between **data complexity, feature dimensionality, and model performance stability**. The overall goal of maximizing Final Test Accuracy while minimizing the Overfitting Gap was best achieved through a combination of **data cleaning** and **sequential dimensionality reduction**.

1. The Impact of Data Cleaning (IQR Capping)

Comparing Experiment 0 (Raw Data) vs. Experiment 1 (IQR Capped):

Metric	Exp 0 (Raw)	Exp 1 (Capped)	Change	Conclusion
Best Accuracy	0.8392 (SVC RBF)	0.8643 (SVC RBF)	+2.51%	IQR Capping significantly improves predictive power.
Best OF Gap	13.50% (SVC Lin)	12.33% (SVC Lin)	-1.17%	Capping slightly reduces variance, stabilizing linear models.
LR Accuracy	0.8141	0.8342	+2.01%	Linear models benefit heavily from outlier management.

Conclusion: Outlier capping (IQR method) is a mandatory preprocessing step for this dataset, leading to an immediate and notable ~2% to 2.5% increase in accuracy across most models without substantially increasing complexity.

2. The Impact of Feature Selection (Reducing Complexity)

The results show that **simplifying the feature space is essential** for managing the critical issue of high variance (overfitting).

A. RFECV (Experiment 2: 67 Features)

Model	Final Test Accuracy	OF Gap (%)
SVC (RBF)	0.8794	13.63%
Logistic Regression	0.8744	12.53%
ANN (MLP)	0.8392	9.88%

- **Key Finding: RFECV** provided the single highest raw accuracy (0.8794) and achieved the lowest gap (9.88%) for the **ANN (MLP)** model, proving it is the most effective feature set for complex classifiers.
- **Dimensionality:** Reducing the features from 243 to 67 significantly improved both linear and non-linear models.

B. Combined RFECV + RF Ranking (Experiment 3: 40 Features)

- **Observation:** Further aggressive reduction from 67 to 40 features using a Random Forest filter **did not consistently improve** performance or robustness. Accuracy peaked at 0.8693 (SVC RBF), slightly lower than the 67-feature set.
- **Critical Gain:** However, the **ANN (MLP)** model achieved its lowest ever overfitting gap of 6.83% with the 40-feature set, demonstrating that aggressive filtering favors simpler, more stable models.

C. PCA (Principal Component Analysis)

- **RFECV + PCA (Experiment 5: 47 Components / 99% Var):** This combination yielded the single **highest accuracy across all experiments**—a peak of 0.8894 with the **SVC (RBF Kernel)**. Crucially, the **OF Gap** for this best-performing model was reduced to **10.46%**, showing a significant improvement in stability compared to the raw data (20.79%).
-

3. Final Model Selection and Robustness

The high **Overfitting Gap** across the board (often >10% even in the best setups) indicates that the N=993 dataset size is highly prone to memorization. Selecting the best model must prioritize **robustness** (high accuracy, low gap).

Rank	Strategy	Model	Final Test Acc	OF Gap (%)	Robustness
1 (Best Accuracy)	Exp 5 (RFECV + PCA)	SVC (RBF Kernel)	0.8894	10.46%	High Acc
2 (Most Robust)	Exp 5 (RFECV + PCA)	SVC (Linear)	0.8593	7.99%	High Stability
3 (Robust ANN)	Exp 3 (RFECV + RF 40)	ANN (MLP)	0.7889	6.83%	Lowest Variance
4 (Best Linear)	Exp 2 (RFECV 67 Feats)	Logistic Regression	0.8744	12.53%	High Acc

Conclusion and Recommended Strategy

The final recommended strategy is the one that combines the best of all worlds: **data cleaning, dimensionality reduction, and advanced modeling.**

- Data Preprocessing:** **IQR Capping** is essential.
- Feature Selection:** The combination of **RFECV → PCA (99% Variance, 47 components)** is the winning method, maximizing information density while removing noise.
- Best Model:** The **SVC (RBF Kernel)** is the best overall performer, achieving a near 89% accuracy.
- Most Reliable Model:** For production where stability is paramount, the **SVC (Linear)** model on the **RFECV + PCA (99%)** features is the most reliable choice, delivering **85.93% accuracy** with the lowest linear gap of **7.99%**.

Further work should focus on reducing the remaining 7–10% overfitting gap through stronger regularization (L2 or L1 penalty) or by collecting more data.

Features 30 Seconds Dataset

The experiments on the **features_30_segs** dataset (1000 samples, 58 original features) demonstrate that the primary challenge is **model stability (variance)**. Achieving better performance relies heavily on **data quality** and **feature selection** to manage high overfitting.

1. The Critical Impact of Data Preprocessing (IQR Capping)

Comparing Experiment 0 (Raw Data) vs. Experiment 1 (IQR Capped):

Metric	Exp 0 (Raw)	Exp 1 (Capped)	Δ Change	Conclusion
Peak Accuracy	0.7900 (ETC)	0.8050 (ETC)	+1.50%	IQR Capping provided a slight but immediate lift in peak predictive performance.
Stability Boost	0.6950 (SVC Lin)	0.7800 (SVC Lin)	+8.50%	The SVC (Linear) model saw the largest test accuracy increase, confirming its high sensitivity to outliers.
OF Gap Reduction	20.40% (Log Reg)	11.96% (Log Reg)	-8.44%	Data cleaning resulted in the most significant reduction in the Overfitting Gap for Logistic Regression, drastically improving its generalization capability.

Key Finding: Initial outlier management (IQR Capping) is **fundamental** for this dataset, reducing overfitting and boosting test accuracy for linear and SVM models by over 8%.

2. Feature Selection: Optimizing for Accuracy and Robustness

Dimensionality reduction was applied to find a balance between the highest accuracy and the lowest variance (OF Gap), moving away from the severe baseline overfitting (≈20%).

Feature Strategy	Final Feats	Best Accuracy	Lowest Overfitting Gap (OF Gap)
Exp 1 (Clean Baseline)	58	0.8050 (ETC)	11.38% (ANN MLP)
Exp 2 (RFECV)	48	0.8200 (ETC)	9.99% (ANN MLP)
Exp 3 (RFECV → RF 40)	40	0.7850 (SVC RBF)	6.40% (Log Reg)
Exp 4 (PCA 90%)	22	0.7750 (SVC RBF)	9.61% (SVC Lin)

- **Accuracy Peak:** The maximum accuracy of 0.8200 was achieved by the **Extra Trees Classifier (ETC)** using the **RFECV-selected 48 features** (Experiment 2).
- **Optimal Stability (Robustness):** The most aggressive feature reduction (**RFECV → RF Top 40**, Experiment 3) yielded the **lowest absolute Overfitting Gap** of **6.40%** with **Logistic Regression**, proving that simpler, highly regularized models benefit most from feature filtering.
- **PCA Efficacy:** PCA (Exp 4) was highly effective at reduction (down to 22 components) while maintaining competitive performance and providing a sub-10% gap for SVC Linear.

3. Final Model Selection and Recommendation

The ideal model provides the best trade-off between **high accuracy** (near 0.8200) and **low variance** (gap ideally <10%).

Rank	Feature Set	Model	Final Test Acc	OF Gap (%)	Primary Value
1 (Peak Accuracy)	Exp 2 (RFECV 48 Feats)	Extra Trees (ETC)	0.8200	18.26%	Highest Predictive Power
2 (Most Robust)	Exp 3 (RFECV → RF 40 Feats)	Logistic Regression	0.7200	6.40%	Maximum Generalization/Stability
3 (Balanced)	Exp 4 (PCA 22 Components)	SVC (Linear)	0.7350	9.61%	Excellent Accuracy/Stability Balance

Conclusion: The **Extra Trees Classifier (ETC)** is the high-risk, high-reward model, achieving the highest accuracy (82.00%) but with a significant 18.26% overfitting gap. For a **stable, deployable solution**, the **SVC (Linear) model** on the **PCA (90%)** features is recommended, offering a solid 73.50% accuracy with a low 9.61% overfitting gap.

Features 3 Seconds Dataset

This is an excellent set of experiments using a significantly larger dataset (9990 rows) derived from 3-second audio segments. The results immediately show a dramatic improvement in model **robustness** compared to the 30-second set.

Here are the exhaustive and summarized conclusions, focusing **only** on the 3-second audio features dataset, structured and presented in English.

The experiments on the **features_3_secs** dataset (9990 samples, 58 features) confirm the hypothesis that **a larger sample size drastically improves model stability and predictive power** in music genre classification. This dataset achieved the highest accuracy and, more importantly, the lowest overall **Overfitting Gap** seen so far.

1. The Impact of Increased Sample Size and Baseline Performance

The move from ≈ 1000 to ≈ 10000 samples completely shifts the analysis focus from **overfitting mitigation** to **accuracy maximization**.

Model	30s (Baseline)	Dataset 3s (Baseline, Exp 0)	Dataset Change in Acc.	Change in OF Gap
KNN	0.6900 Acc / 32.01% Gap	0.9199 Acc / 10.12% Gap	+22.99%	-21.89%
Log Reg	0.7100 Acc / 20.40% Gap	0.7362 Acc / 2.50% Gap	+2.62%	-17.90%
ETC	0.7900 Acc / 20.26% Gap	0.8964 Acc / 11.33% Gap	+10.64%	-8.93%

Key Finding: The larger sample size is the single most important factor. It transformed the **K-Nearest Neighbors (KNN)** model into the **top performer**, achieving an astonishing **91.99% accuracy** while simultaneously reducing its high variance by over 20 percentage points.

2. The Effect of Preprocessing and Dimensionality Reduction

Given the low baseline variance, cleaning and feature reduction were evaluated for their ability to push accuracy even higher or achieve near-perfect stability.

A. Data Cleaning (Experiment 1: IQR Capped)

- Observation:** IQR Capping had **negligible impact** on this dataset. The best accuracy (KNN) moved from 0.9199 to 0.9209 (a minor +0.10% gain), and the Overfitting Gap for Logistic Regression barely moved (2.50% to 2.52%).
- Conclusion:** With ample data (N=9990), the models are far less susceptible to individual outlier noise, making aggressive data cleaning less critical than it was for the smaller datasets.

B. Principal Component Analysis (Experiment 3: PCA 90%)

- Observation:** Reducing the feature count from 58 to 30 components drastically **hurt predictive accuracy** across the board. The peak accuracy dropped from 0.9199 to 0.8559 (a -6.40% loss).
- Robustness Trade-off:** While PCA provided the **absolute lowest Overfitting Gap** for Logistic Regression at **1.33%**, this stability came at an unacceptable cost to predictive power. It suggests the original 58 features contain relevant information that is not fully captured by the first 30 principal components.

3. Final Model Selection and Robustness (The Time Factor)

Since RF and SVC were excluded, focus on the most effective fast-training models.

Rank	Strategy	Model	Final Test Acc	OF Gap (%)	Primary Value
1 (Peak Performance)	Exp 0/1 (58 Feats)	K-Nearest Neighbors	0.9209	9.47%	Highest Accuracy
2 (Best Stability)	Exp 3 (PCA 30 Comp.)	Logistic Regression	0.6682	1.33%	Lowest Variance
3 (Ensemble Power)	Exp 0 (58 Feats)	Extra Trees (ETC)	0.8964	11.33%	Strong, Fast Non-Linearity

Conclusion

The definitive conclusion is that the **large sample size** provided by the 3-second segments (9990 total) solves the overfitting problem.

1. **Optimal Features:** The **full set of 58 features** should be retained. PCA, while reducing variance, incurs too great a penalty on accuracy.
2. **Best Model:** The **K-Nearest Neighbors (KNN)** model is the undisputed champion, achieving 92.09% accuracy, a phenomenal result that mirrors human capacity for short audio classification.
3. **Most Robust:** For applications where even 10% variance is too high, the **Logistic Regression** model is incredibly robust with only a **2.50%** gap on the full feature set.

Ranking of 10 Best Models & Data

```
best_model_summary, ranking_df = find_best_model_global(list_of_dfs, experiment_names)
ranking_df.sort_values(by='Final Test Accuracy', ascending=False).head(10)
```

--- Global Best Model Found ---
Model: K-Nearest Neighbors
Feature Set: results_3_1
Final Test Accuracy: 0.920921
Overfitting Gap (%): 9.467638
Name: 117, dtype: object

	Model	Optimal Hyperparams	CV Mean Train Acc	CV Mean Test Acc	Overfitting Gap (%)	Final Test Accuracy	Feature Set
117	K-Nearest Neighbors	{'classifier__n_neighbors': 3, 'classifier__we...	0.999333	0.904656	9.467638	0.920921	results_3_1
112	K-Nearest Neighbors	{'classifier__n_neighbors': 3, 'classifier__we...	0.999333	0.898151	10.118185	0.919920	results_3_0
118	Extra Trees (ETC)	Default	0.999333	0.882257	11.707563	0.903403	results_3_1
113	Extra Trees (ETC)	Default	0.999333	0.880630	11.870313	0.901401	results_3_0
40	SVC (RBF Kernel)	{'classifier__C': 1, 'classifier__gamma': 0.01...	0.959699	0.855127	10.457195	0.889447	results_5
16	SVC (RBF Kernel)	{'classifier__C': 10, 'classifier__gamma': 'sc...	0.999020	0.862722	13.629887	0.879397	results_2
17	Logistic Regression	{'classifier__C': 1, 'classifier__penalty': 'l2'}	0.982928	0.857674	12.525381	0.874372	results_2
24	SVC (RBF Kernel)	{'classifier__C': 10, 'classifier__gamma': 'sc...	0.997481	0.840127	15.735476	0.869347	results_3
18	SVC (Linear)	{'classifier__C': 0.1, 'classifier__kernel': '...	0.953399	0.841329	11.207004	0.864322	results_2
25	SVC (Linear)	{'classifier__C': 0.1, 'classifier__kernel': '...	0.912679	0.816060	9.661876	0.864322	results_3

Best Model & Data

- Data: Task 3 (larger data over features avoids overfitting).
- Model: KNN (number of neighbours = 3, weights = distance)

```
pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('classifier', KNeighborsClassifier(n_neighbors=3, weights="distance"))
])
```

```
--- CLASSIFICATION REPORT ---
              precision    recall  f1-score   support

    blues      0.9909      0.9810      0.9859      1000
  classical      0.9841      0.9940      0.9890       998
    country      0.9708      0.9679      0.9694       997
     disco      0.9812      0.9900      0.9856       999
    hiphop      0.9889      0.9800      0.9844       998
     jazz      0.9840      0.9820      0.9830      1000
     metal      0.9900      0.9950      0.9925      1000
       pop      0.9890      0.9910      0.9900      1000
    reggae      0.9743      0.9870      0.9806      1000
       rock      0.9817      0.9669      0.9743       998

 accuracy      0.9835
macro avg      0.9835      0.9835      0.9835      9990
weighted avg      0.9835      0.9835      0.9835      9990
```

Confusion Matrix for Best Global Model

True Label	blues	981	0	5	1	3	2	0	0	4	4
	classical	0	992	0	0	0	5	0	1	0	0
	country	7	2	965	5	0	6	0	2	7	3
	disco	0	2	2	989	0	0	1	3	0	2
	hiphop	0	0	2	2	978	2	0	4	8	2
	jazz	1	9	7	0	0	982	0	0	1	0
	metal	0	0	0	2	0	0	995	0	0	3
	pop	0	0	3	3	1	1	0	991	0	1
	reggae	0	1	1	3	4	0	0	1	987	3
	rock	1	2	9	3	3	0	9	0	6	965
			blues	classical	country	disco	hiphop	jazz	metal	pop	reggae